

Day 21 (Part 3): PortFast (STP Toolkit)



PortFast on trunk ports



- The standard PortFast configuration commands only enable PortFast on access ports.
 - Per-port: SW1(config-if)# **spanning-tree portfast**
 - Default: SW1(config)# **spanning-tree portfast default**
- In some cases, you might want to enable PortFast on a trunk port:
 - A port connected to a virtualization server with virtual machines (VMs) in different VLANs.
 - A port connected to a router via router-on-a-stick (ROAS).
- This can only be configured per-port in interface config mode:
 - SW1(config-if)# **spanning-tree portfast trunk**

```
SW1(config-if)# spanning-tree portfast trunk
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION
```

```
SW1# show spanning-tree interface g0/1 detail
Output omitted
The port is in the portfast edge trunk mode
Output omitted
```



PortFast edge



- In modern Cisco switches, if you use the commands covered in this lecture, the device will automatically add the **edge** keyword to the configuration.
 - SW1(config-if)# **spanning-tree portfast**
 - In the running-config: **spanning-tree portfast edge**
- SW1(config-if)# **spanning-tree portfast trunk**
- In the running-config: **spanning-tree portfast edge trunk**
- SW1(config)# **spanning-tree portfast default**
- In the running-config: **spanning-tree portfast edge default**

- You can use either version of the commands when configuring PortFast.
- The end result is the same: **edge** will always be added in the configuration.
- spanning-tree portfast disable** doesn't use the **edge** keyword.

```
SW1(config)# interface g0/1
SW1(config-if)# spanning-tree portfast

SW1# show running-config interface g0/1
Building configuration...
```

```
Current configuration : 113 bytes
!
interface GigabitEthernet0/1
 switchport access vlan 10
 switchport mode access
 spanning-tree portfast edge
end
```

- Use **show running-config interface interface-name** to view the running-config for the specified interface.
- Doesn't work in Packet Tracer.

Automatically added **edge** to the end of the command.





Summary

- When a host connects to a switch port, by default it takes 30 seconds before the port can send/receive data.
- **PortFast** allows a switch port to immediately enter the STP **Forwarding** state, bypassing the **Listening** and **Learning** states.
- It can be configured in two ways:
 1. Interface config mode:
SW1(config-if)# **spanning-tree portfast [edge]**
This enables PortFast only on the individual interface. Only active when in access mode.
 2. Global config mode:
SW1(config)# **spanning-tree portfast [edge] default**
This enables PortFast on all access ports.
You can then use **spanning-tree portfast disable** to disable PortFast on specific ports.
- PortFast should not be configured on ports connected to a switch, as it can cause temporary Layer 2 loops.
- You can enable PortFast on a trunk port with **spanning-tree portfast [edge] trunk**.